

Annie Prasanna Manoharan

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Education

University of Arkansas

Aug 2024 – May 2026

Master of Science in Computer Engineering

– Coursework: Artificial Intelligence, Machine Learning, Privacy Enhancing Techniques

Sri Ramakrishna Engineering College

Aug 2019 – May 2023

Bachelor of Engineering in Electronics and Communication

India

Technical Skills

Languages: Python, SQL, C++, C#, Embedded C, Java, Verilog

Data Engineering: Apache Kafka, Apache Flink, Apache Airflow, Databricks, Apache Spark, AWS, Flink SQL, ClickHouse, PySpark

Machine Learning: PyTorch, Scikit-learn, Computer Vision, LangChain, RAG, LSTM, YOLO11n-Pose, OC-SORT, FAISS

Cloud and Tools: AWS, Glue, Lambda, EMR, S3, Grafana, Docker, Git, Flask

Embedded and Test: LabVIEW, NI TestStand, MAVIS, Test Automation, FPGA, Microblaze, PIC16F887, Ansys HFSS

Projects

Real-Time Medical IoT Analytics Platform

Apache Kafka, Apache Flink, Flink SQL, Python, ClickHouse, Grafana

- Developed an end-to-end, sub-second latency streaming platform using Apache Flink SQL and Apache Kafka to ingest, process, and score high-frequency medical IoT device data
- Designed high-performance ClickHouse schemas using the MergeTree engine sorted by composite keys to physically align disk layouts with clinical querying patterns, isolating ingestion via an optimized Python multi-topic batch consumer
- Implemented temporal Flink OVER windows partitioned by device ID to dynamically compute rolling Z-scores over 5-second sliding intervals, applying custom watermark delay strategies to cleanly handle out-of-order data streams

Local LLM-Powered News Research Platform

- Python, LangChain, Streamlit, FAISS, Ollama, nomic-embed-text, Llama 3.2, WebBaseLoader
- RAG pipeline via LangChain orchestrating semantic queries against vectorized data using local Llama 3.2 instances
 - End-to-end local ETL pipeline using Streamlit and WebBaseLoader to ingest, clean, chunk, and preprocess unstructured web content on demand

Resume Classification Model

Python, Scikit-learn, Pandas, TF-IDF, Logistic Regression, Flask

- Deployed through a Flask REST API for real-time prediction
- End-to-end NLP classification pipeline: TF-IDF feature extraction + Logistic Regression across 25 job-role categories

Professional Experience

Data Engineering Intern

Jul 2024 – Present

InsOps Inc.

Chicago, Illinois

- Built data workflows integrating SQL Server, Azure SQL, AWS S3, and cloud databases, preparing structured datasets for downstream data science and analytics
- Developed data pipelines in AWS using AWS Glue, Lambda, EMR, SQL, and Apache Airflow for orchestration, with S3 cloud storage, for scalable data ingestion, transformation, validation, and analytics
- Worked with distributed data processing and lakehouse concepts, including scalable ETL/ELT, cloud data storage, SQL transformations, and pipeline orchestration

Project Engineer

Soliton Technologies

India

- Developed customized software solutions for Medtronic as the sole developer, automating approximately 94% of manual test cases while meeting medical-device quality and validation requirements — see Q4 before using this figure
- Integrated automated testing and validation workflows to improve test coverage, reduce manual execution, and support repeatable software verification

Intern

Sep 2021 – Oct 2021

National Institute of Technology, India

- Conducted research on antenna design incorporating Substrate Integrated Waveguide (SIW) and deep-well cavity structures to enhance antenna performance
- Designed and modeled antenna structures using Ansys HFSS for electromagnetic simulation and performance analysis